Research Summary Sheet - Article #1

Complete the following form for each artifact that validates your problem statement.

Source (APA format and include a link to the source)

Cargo spacecraft | EBSCO. (2024). EBSCO Information Services, Inc. | Www.Ebsco.Com.

https://www.ebsco.com/research-starters/engineering/cargo-spacecraft

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Artifact Summary

It defines what cargo spacecrafts are: they are designed to carry supplies into orbit. Cargo Spacecraft was first made by scientists who worked for the Soviet Union in the 1960s. They created the TKS which was a Transport Supply Spacecraft, but this required a pilot and crew. In order to maintain a permanent crew however, ISS needed a steady stream of supplies going in and out. The European Agency later created the Automated Transfer Vehicle which could carry around 7600 kg and the Japan Aerospace Exploration Agency developed the H-II Transport Vehicle which could carry around 6,000 kg of material. The ATV and HTV were not manned autonomous/remote control and launched from earth. After the space shuttle retired, and SpaceX launched the Dragon which was the first reusable cargo space craft. Engineers are continuing to improve more designs.

Artifact Critique

This artifact talks about the differences between the cargo shuttles that were developed over the years, but doesn't mention the critiques regarding any of the space shuttles. I would like to learn more about how the Dragon could further be improved and what scientists are looking for in future spacecrafts. As of right now, it seems that most of these shuttles are launched from Earth then attached themselves to the ISS and humans have to retrieve the cargo therefore an autonomous cargo transport vehicle between the supply building and the payloader seems to be what engineers are trying to build.

Reliability

The article has referenced around 10 articles all sourced from recent NASA articles, and the articles itself was published in around 2020. With all this information I believe that the source is reliable as it uses direct research from the NASA inventory and is a factual rather than biased article.